

Features of Python

Python is one of the most popular programming languages because of its simplicity and versatility. Its key features are:

1. **Easy to Learn and Use** – Python has a simple and readable syntax similar to English.
2. **Interpreted Language** – Python code is executed line by line, making debugging easier.
3. **Dynamically Typed** – Variables do not require explicit type declarations.
4. **Object-Oriented** – Supports classes, objects, inheritance, polymorphism, and encapsulation.
5. **Platform Independent** – Python programs can run on Windows, Linux, and macOS without modification.
6. **Open Source** – Python is free to use and has a large community of contributors.
7. **Extensive Standard Library** – Provides built-in modules for various tasks such as file handling, networking, and data processing.
8. **High-Level Language** – Hides hardware-level details, making programming easier.
9. **Supports Multiple Programming Paradigms** – Includes procedural, object-oriented, and functional programming.
10. **Large Community Support** – Extensive documentation, tutorials, and third-party packages are available.
11. **Automatic Memory Management** – Python manages memory automatically using garbage collection.
12. **Extensible and Embeddable** – Can be integrated with languages like C and C++.